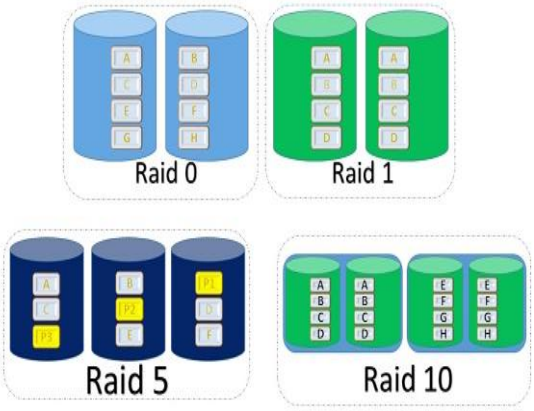


CLOUD COMPUTING COURSE

Lecture22- Cloud Storage (5)

ABSTRACT
Lecture22 Introduces RAID variations.

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Centralized Storage

RAID Types (Variations)

RAID is an umbrella name for different variations of combining hard-disks that would then appear to your computer as one single hard-disk.

Multiple disks are combined with different goals in mind, hence the different variations. Goals can be **Faster access to data**, more **storage space** and/or **higher reliability.**

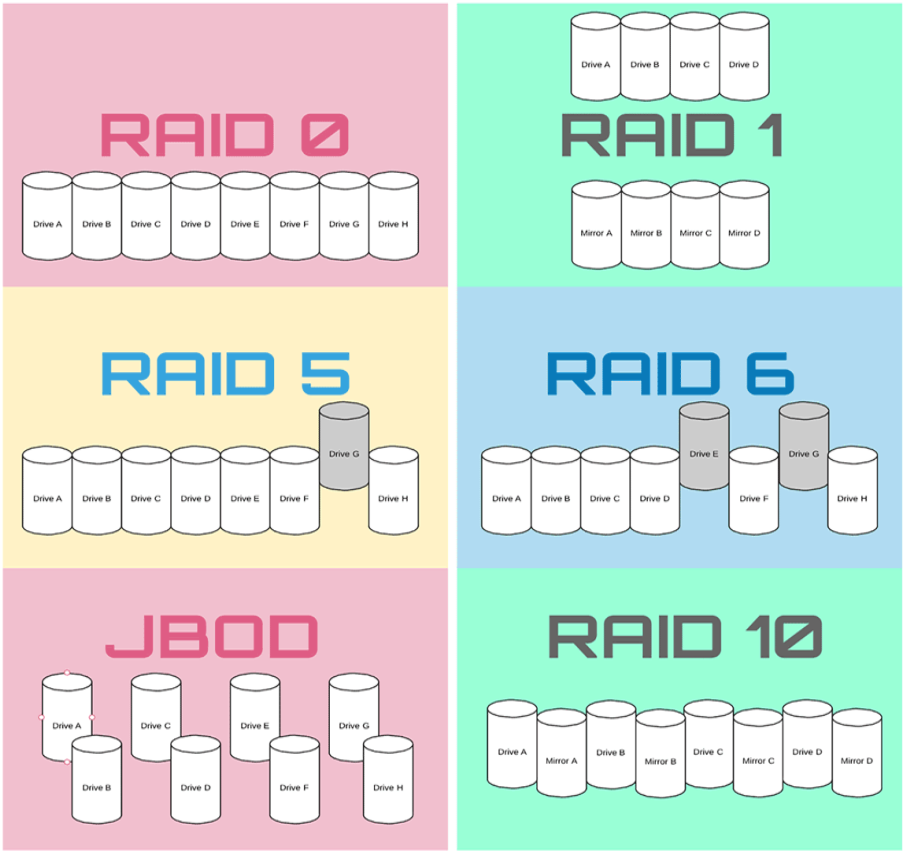
These variations are named "**RAID**" followed by a **number**. For example **RAID 0**, **RAID 1**, etc. Each fitting for a particular purpose.

The different variations come with different levels of:

Speed, **reliability**, **availability** and the **minimum number of hard-drives** needed.

RAID Variations (Types)

- RAID 0 – Striping
- RAID 1 – Mirroring
- RAID 2 – Bit level Striping
- RAID 3 / RAID 4 – Striping with Parity
- RAID 5 – Striping with rotating parity
- RAID 6 – Striping with double rotating parity
- Nested RAID (RAID 10)



RAID 0 – Striping

In RAID 0 data is written in **parallel** to 2 or more disks. Think of it as writing block 1 to disk 1, block 2 to disk 2, block 3 to disk 3 etc.

Minimum number of required drives: 2 drives (even number).

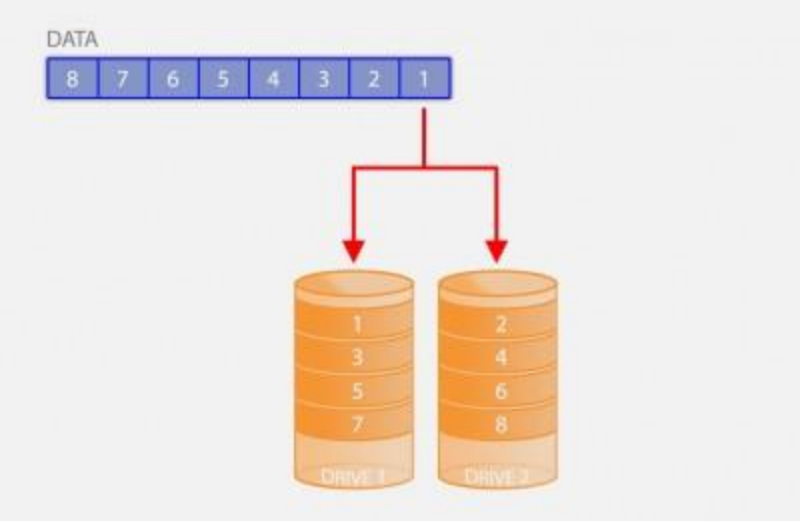
Space efficiency: 100% of the sum of the capacity of the drives.

Positives:

- The great advantage is **SPEED** as data is written or read in parallel.
- Another advantage is that all of the **storage capacity** will be available for storage.

Negatives:

The big downside is **lack of reliability**: If one disk crashes all data is gone, as half of the information will be lost. RAID 0 is not fault-tolerant.



RAID 0 – Striping

Ideal use

RAID 0 is ideal for non-critical storage of data that have to be read/written at a high speed, such as on an image retouching or video editing station.

RAID 1 – Mirroring

RAID 1 arrays mirrored data of one drive to another drive. Therefore, when your computer writes data to the array, the controller will actually write the data twice: to disk 1 and to disk 2.

If a drive fails, the controller uses either the data drive or the mirror drive for data recovery and continues operation.

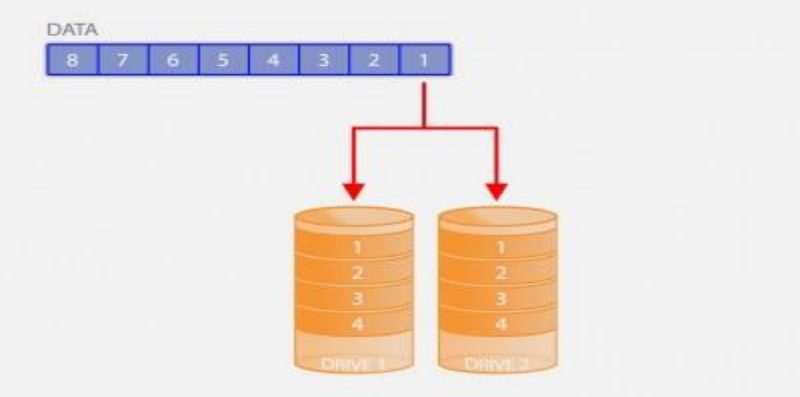
Minimum number of required drives: 2 drives (even number).

Space efficiency: 50% of the sum of the capacity of the drives.

Positives:

- Increased **reliability** since data is always saved in duplicate on two different drives. If one drive dies, the other will still be able to provide you the data. In case a drive fails, data do not have to be rebuild, they just have to be copied to the replacement drive.
- RAID 1 is a very simple technology.

Negatives:
The main disadvantage is that the effective storage capacity is only half of the total drive capacity because all data get written twice.



RAID 1 – Mirroring

Ideal use

RAID-1 is ideal for mission critical storage, for instance for **accounting systems**.

RAID 5 – Striping with rotating Parity

- RAID 5 is the most common secure RAID level.
- It requires at least 3 drives but can work with up to 16.
- Data blocks are striped across the drives and on one drive a parity checksum of all the block data is written.
- The parity data are not written to a fixed drive, they are spread across all drives, as the drawing below shows.
- Using the parity data, the computer can recalculate the data of one of the other data blocks, should those data no longer be available.
- That means a RAID 5 array can withstand a single drive failure without losing data.
- Although RAID 5 can be achieved in software, a hardware controller is recommended. Often extra cache memory is used on these controllers to improve the write performance.

Minimum number of required drives: 3 drives,

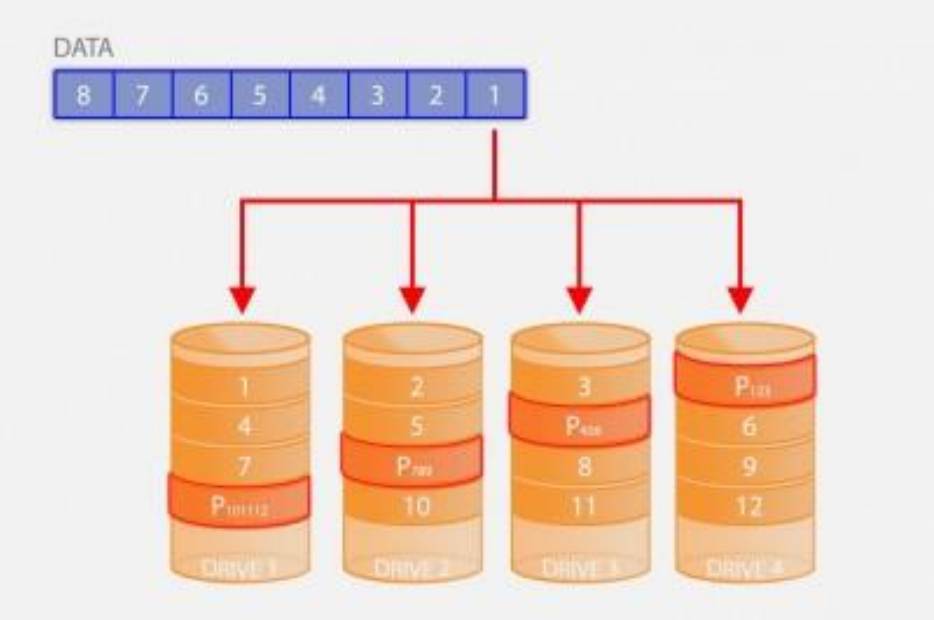
Space efficiency: **Sum of the capacity of all drives – (Sum of the capacity of all drives / number of drives)**

Advantages:

- Good **performance**: Read data transactions are very fast while write data transactions are somewhat slower (due to the parity that has to be calculated).
- Very good data **reliability**: If a drive fails, you still have access to all data, even while the failed drive is being replaced and the storage controller rebuilds the data on the new drive.

Disadvantage:

This is complex technology. If one of the disks in an array using 4TB disks fails and is replaced, restoring the data (the rebuild time) may take a day or longer, depending on the load on the array and the speed of the controller. If another disk goes bad during that time, data are lost forever.



RAID 5 – Striping with distributed Parity

Ideal use

RAID 5 is a good all-round system that combines efficient storage with excellent security and decent performance. It is ideal for file and application servers that have a limited number of data drives.

RAID 6 – Striping and Double rotating Parity

RAID 6 can be seen as an extended RAID 5. It actually maintains 2 parity calculations which are written to two drive, and like RAID 5 these are equally distributed offering the ability to even recover from 2 failing drives.

Minimum number of required drives: 4 drives.

Space efficiency:

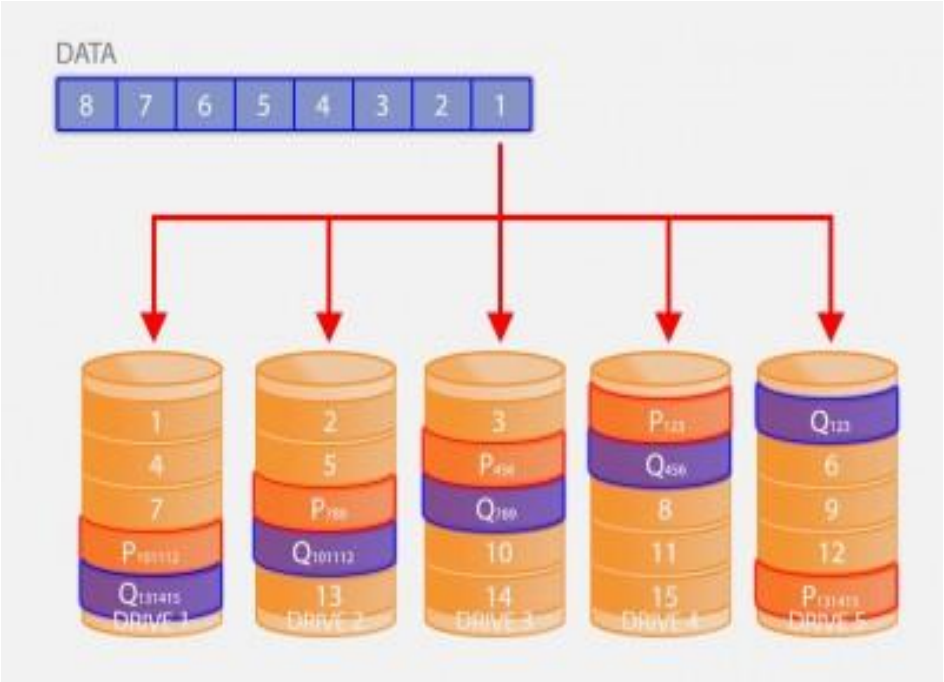
Sum of the capacity of all drives – (2 x Sum of the capacity of all drives / number of drives)

Advantages

- Good **speed** performance: Like with RAID 5, read data transactions are very fast.
- Higher **reliability**: If two drives fail, you still have access to all data, even while the failed drives are being replaced. Therefore, RAID 6 is more secure than RAID5.

Disadvantages

- Write data transactions are slower than RAID 5 due to the additional parity data that have to be calculated.
- The remaining **capacity** will be lower than RAID 5.



RAID 6 – Striping with double distributed parity

Ideal use

RAID 6 is a good all-round system that combines efficient storage with excellent security and decent performance. It is preferable over RAID 5 in file and application servers that **use many large drives** for data storage.

Nested RAID configurations

Certain RAID variations can be combined, depending on your RAID controller of course, and are referred to as "Nested RAID".

Below an example of **RAID 10** or RAID 1 + 0 … Data is written in parallel (RAID 0) and then written mirrored (RAID 1).

It is possible to combine the advantages (and disadvantages) of RAID 0 and RAID 1 in one single system. This is a nested or hybrid RAID configuration. It provides security by mirroring all data on secondary drives while using striping across each set of drives to speed up data transfers.

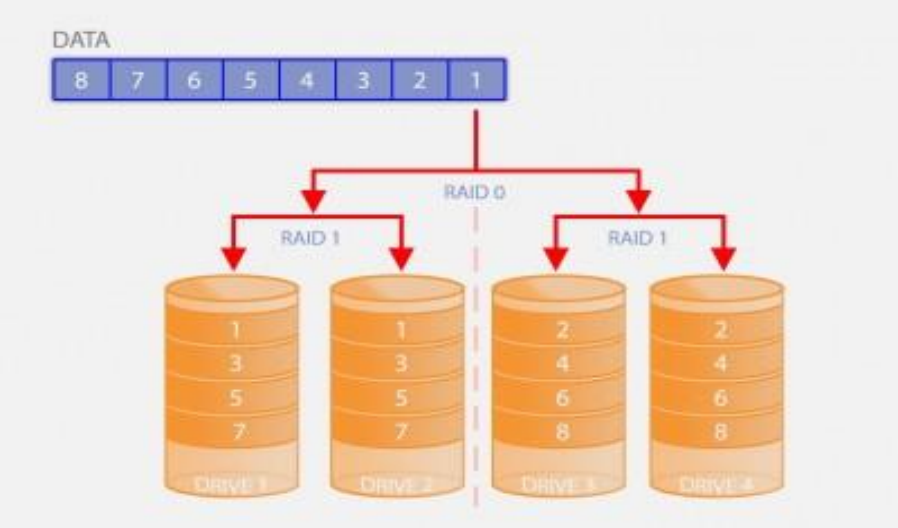
Other variations are **RAID 0+1**, **RAID 50** (RAID 5 + RAID 0), **RAID 100** (RAID 1 + RAID 0 + RAID 0), **RAID 53** (RAID 5 + RAID 3) and **RAID 60** (RAID 6 + RAID 0).

Positives

Custom RAID. You can configure your own fitting your own needs.

Negatives

Not all RAID controllers support this and the resulting **storage capacity** will be much lower than other configurations.



RAID 10 – Nested RAID

Related videos

RAID Types

https://youtu.be/U-OCdTeZLac

RAID5 VS RAID6

https://youtu.be/UuUgfCvt9-Q